THE BASE CHANGE MAP FOR THE Q-TH DIRECT IMAGE IS AN ISOMORPHISM IF AND ONLY IF THE TWO STRICT HENSELISATIONS HAVE THE SAME COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the rewriting of the fibre product, the identification of the map in (1) with the stalk of the map in (2), and the concluding stalk criterion.

Editorial note. The label lemma-base-change-f-star-general-stalks names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement's displayed diagram is the author's and is delivered verbatim; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Stalks of pushforwards as cohomology of the fibre product. Same chapter, the theorem describing the stalk of a pushforward as the cohomology of the corresponding base change, together with the stalk-pullback lemma. Supplies the identification of the map in condition (2) with the map in condition (1).
- A map of sheaves is an isomorphism if and only if it is so on stalks. Same chapter, the exactness-on-stalks theorem for the etale topology. Supplies the final equivalence.

Source reference. Original reference 008H: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4649–4649. Prerequisite pointer invoked at proof line(s) 16527; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Prerequisite pointer invoked at proof line(s) 16528; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03Q9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8015–8015. Prerequisite pointer invoked at proof

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line(s) 16526; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Consider a cartesian diagram of schemes*

$$\begin{array}{ccc} X & \xleftarrow{h} & Y \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle e} \\ S & \xleftarrow{g} & T \end{array}$$

where $g : T \to S$ is quasi-compact and quasi-separated. Let $\mathcal{F}$ be an abelian sheaf on $T_{étale}$. Let $q \geq 0$. The following are equivalent

(1) *For every geometric point $\overline{x}$ of X with image $\overline{s} = f(\overline{x})$ we have*
$$H^q(\mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \times_S T, \mathcal{F}) = H^q(\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}}) \times_S T, \mathcal{F})$$
(2) *$f^{-1}R^q g_*\mathcal{F} \to R^q h_* e^{-1}\mathcal{F}$ is an isomorphism.*

Proof. Since $Y = X \times_S T$ we have $\mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \times_X Y = \mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) \times_S T$. Thus the map in (1) is the map of stalks at $\overline{x}$ for the map in (2) by Theorem 03Q9 (and Lemma 008H). Thus the result by Theorem 03PU. □